ScienceWorld: environment inspection report

Wave-0 M5, CA-Benchmark project

2026-07-23

Scope. Hands-on inspection of the ScienceWorld text environment (rung 4 of the credit-assignment benchmark): what the tasks are, how they are scored and evaluated, how many trials exist, rollout speed on our node, the sandbox/action surface, a full worked trajectory, expert-data availability, published baselines, and the challenges that bear on training. All local numbers are reproduced by `scripts/scienceworld/audit_structure.py`, `analyze_audit.py`, and `bench_and_probe.py`; raw assets are in `assets/`. Package: `scienceworld 1.2.3` (pip) on OpenJDK 17, one CPU worker.

1 What the environment is (the sandbox)

ScienceWorld (Wang et al., EMNLP 2022; arXiv 2203.07540) is a **text-based interactive simulator of elementary-school science**. The agent navigates a map of **10 locations** centered on a house (hallway, kitchen, bathroom, workshop, art studio, greenhouse, bedroom, living room, foundry, outside/ garden), populated with **up to 200 object types**, carrying out science procedures. It is **partially observable**: `look around` reveals only the current room; the underlying engine tracks a full thermodynamic / chemical / biological state (temperatures, states of matter, electrical circuits, plant and animal life-cycles) that the agent must actively probe with instruments.

Engine & interface. The simulator core is **∼40k lines of Scala** (compiled to a JAR, run on Java 1.8+) exposed to Python via `py4j` — a JVM must be present (we installed OpenJDK 17). Verified from the source: each `ScienceWorldEnv` launches **its own dedicated JVM process** (plus a Python callback server); there is no documented thread-safety mechanism, so **parallelism is process-level** (the paper's RL runs used 8 parallel env instances). For us this means the environment is CPU/JVM-bound and single-threaded per instance; worker throughput scales with the number of env processes we co-locate with the trainer, not with threads.

Action space (25 templates). Actions are *template + object(s)* combinations:

```
activate/deactivate OBJ   open/close OBJ      focus on OBJ
go OBJ                    teleport OBJ        look around / look at / look in OBJ
pick up / put down OBJ    move OBJ to OBJ     pour OBJ in OBJ
dunk OBJ in OBJ           mix OBJ             use OBJ on OBJ
read OBJ                  eat OBJ             flush OBJ
wait / wait1              inventory           task
```

The **combinatorial action space** is the headline difficulty: the paper estimates ≈**200,000 naively-possible action–object combinations per step**; in our own probe a single `boil` start-state already exposes **115 valid combinations**, growing as the agent uncovers objects. A free-text agent (our ReAct policy) is *not* handed this list — it must generate a valid `Thought:/Action:` string, and malformed or inapplicable actions get a soft error observation (e.g. `"You can't move something into itself."`). Critically, the paper reports that all but two of its baselines relied on a **valid-action-detection aid** at test time to shrink this search — a crutch our generative agent does not use, which makes valid-action-rate a first-class diagnostic. **Simplifications** can be toggled; the audit used the standard `easy` set (`teleportAction`, `openDoors`, `selfWateringFlowerPots`, `noElectricalAction`) — these must be frozen identically across all arms and recorded, since they change both horizon and answerability.

2 The tasks and how they are scored (the evaluation)

There are **30 tasks** (10 topic categories) spanning thermodynamics, chemistry/mixing, electricity, classification, biology (life stages, lifespans, plant growth, genetics), and measurement; we group them into **10 families** (`assets/families.csv`). The paper reports **7,200 total variations across all 30 tasks** — different target substances, objects, and layouts. Each task ships fixed **train / dev / test** variation splits (≈50/25/25%), so "unseen variations" is a built-in generalization split — no custom splitting needed. The original protocol resets an episode at success, failure, or a **100-step cap**; we replace that single global cap with per-family caps (below), since horizons range ∼40×.

Scoring is a dense, cumulative 0–100 subgoal score. The per-step `reward` returned by the API is the score *delta*. From replaying 90 gold trajectories (3 train variations × 30 tasks):

- All 90 gold paths reach exactly **score 100 with `done=True`**; no negative deltas on-path.
- Subgoal firing density: median **0.27** score-increments per step (range 0.05–0.62) — dense relative to a terminal-only signal, but far from every-step.

- **Magnitudes are heterogeneous and task-specific: 42 distinct values from 1 to 72.** The oracle is *not* a uniform unit-per-subgoal reward. Any process-reward (PRM) arm in the side experiments must match this granularity or renormalize explicitly and say so.
- **Off-gold failure is catastrophic:** a wrong `focus on` (focusing on the wrong object) returns **reward** -100 **and immediately terminates the episode**. `focus` is thus both a hard subgoal gate and an instant-death / reward-hacking surface for exploration.

Our reward convention (proposed, confirm at W1 manifest). Core-matrix arms train on *terminal success only*: success $\Leftrightarrow$ final score = 100; a -100 focus-death maps to terminal reward 0 and episode end (mirroring ALFWorld failure, keeping terminal reward in $\{0, 1\}$). The dense 0–100 oracle is reserved for the PRM-ladder / redistribution side experiments, where the -100 event is exactly the kind of asymmetry the redistribution-vs-separation question probes.

Oracle isolation (checked). Observation text on all 90 gold paths contained zero `score`/`reward`/`subgoal`/`points` strings; the score lives only in the API `info` dict, and the `task` command exposes only the task description (already in the prompt). So a transcript-only agent is blind to the oracle — a prerequisite for the "rubric PRM must not see oracle state" guarantee. (One residual: audit our own env *wrapper* once written.)

3 How many trials, and rollout speed

Trials. 30 tasks with widely varying variation counts — from **4** train variations (`identify-life-stages-2`) to **692** (`inclined-plane-friction-named-surfaces`). The `find-*` and `test-conductivity` families have 150–450 train variations each. The paper's **7,200 total variations** (train+dev+test across all tasks) means that, unlike a fixed-size QA set, ScienceWorld supplies a large pool of fresh task instances — the training-time question is the *sampling* policy (uniform-per-task vs uniform-per-variation), which we must fix before Wave 1 because the 4-vs-692 imbalance changes what one "epoch" means and which tasks dominate the gradient.

Rollout speed (single JVM env, our node; `assets/speed_bench.json`):

Operation	measured
`load(task,var)` + `reset`	0.22 s/episode (0.15–0.36)
Gold-action stepping	**13.8 steps/s** (73 ms/step)
Random stepping, `step` only	15.3 steps/s (65 ms/step)
Random stepping incl. `get_valid_actions` each step	9.0 steps/s (112 ms/step)

Reading for training budget. At ≈65–75 ms/env-step, one env instance yields ∼14 steps/s. The horizon is the multiplier that matters: a 30-step task costs ∼2 s of pure env time, but a 200-step `inclined-plane` episode costs ∼14 s *of CPU env time alone*, before any LLM generation. This is why ScienceWorld is the most expensive rung per environment-episode and why long-horizon families dominate wall-clock. Querying the valid-action set every step (which we do *not* need for a free-text agent) nearly halves throughput — keep it out of the hot loop. Env parallelism is process-level (one JVM each), so worker-level throughput scales with the number of env workers we run alongside the trainer.

4 A full trajectory (gold/expert)

Expert data is available for free: the engine generates a **gold action sequence** for any task/variation (`generateGoldPath=True` → `get_gold_action_sequence()`). This is our rung-4 SFT/π_{base} source (rendered into the ReAct grammar, with the score channel stripped — oracle isolation). No external expert dataset is needed. Below is a complete gold trajectory for a short task; green rows mark subgoal firings (with the reward increment), showing how the dense signal is distributed along the path.

Task `find-animal` (variation 0), 10 gold steps, final score 100.
"Your task is to find a(n) animal. First, focus on the thing. Then, move it to the red box in the kitchen."

t	action	observation	Δscore
0	`open door to greenhouse`	The door is already open.	+8
1	`go to greenhouse`	You move to the greenhouse.	+9
2	`open door to outside`	The door is already open.	
3	`go to outside`	You move to the outside.	
4	`look around`	…an axe, a blue jay egg, …(many objects)	
5	`focus on blue jay egg`	You focus on the blue jay egg.	+50
6	`pick up blue jay egg`	You move the blue jay to the inventory.	+8
7	`open door to kitchen`	The door is already open.	
8	`go to kitchen`	You move to the kitchen.	+8
9	`move blue jay egg to red box`	You move the blue jay to the red box.	+17

A second, richer trajectory (`change-the-state-of-matter-of` water, 22 steps, in `assets/full_traj_*.json`) shows the harder physics: the agent picks up a thermometer, puts water in a metal pot, `focus on substance in metal` `pot` (**+66**, the big subgoal gate), moves it to the freezer (+2), then `wait`s while probing temperature (10 °C → 3 °C) until the water freezes and the final `wait` fires **+29** to reach 100. It illustrates three training-relevant traits at once: **partial observability** (state read only via the thermometer), **delayed physical dynamics** (must `wait` for freezing — reward arrives many steps after the causal action), and **a dominant single subgoal** (`focus` worth 66/100), which shapes what credit assignment has to discover.

5 Published baselines (original paper + follow-ups)

Original paper (Table 2; test variations; 0–1 scale). The headline finding is that *elementary science is hard for text agents and larger models are not better*: a 1.5M-parameter RL agent trained interactively for 100k steps beats an 11B model trained statically for science QA.

Agent	Params	Avg score (0–1)
Random-valid	—	0.03
DRRN (best baseline)	1.5M	**0.17**
KG-A2C	5.5M	0.11
CALM (GPT-2)	131M	0.05
Behavior Cloning (T5-11B)	11B	0.08
Text Decision Transformer (T5-11B)	11B	0.08

Best agent 0.17/1.0 — comparable to text agents on medium-difficulty interactive-fiction games. The paper notes all but two baselines leaned on the valid-action aid at test time. Source: arXiv 2203.07540 Table 2.

Follow-ups (0–100 scale; note the differing eval sets — not directly comparable across rows). Modern LLM agents changed the picture dramatically:

Method	Base model	Eval set	Avg score
DRRN / KG-A2C / CALM	(paper baselines)	270 test vars	15.6 / 11.4 / 5.1
SayCan	GPT-4	270 test vars	33.8
ReAct	GPT-4	270 test vars	36.4
Reflexion	GPT-4	270 test vars	45.3
SwiftSage	T5-large (770M) + GPT-4	270 test vars	**84.7**
PPO	Llama-2-7B-Chat	194 seen / 241 unseen	59.4 / 51.7
SFT	Llama-2-7B-Chat	194 seen / 241 unseen	67.4 / 53.0
ETO	Llama-2-7B-Chat	194 seen / 241 unseen	**73.8 / 65.0**
GPT-4 (ETO paper)	GPT-4	194 seen / 241 unseen	64.8 / 64.4
BC-large	Llama-2-7B-Chat	AgentGym split	74.5
AgentEvol	Llama-2-7B-Chat	AgentGym split	38.0
GPT-4-Turbo (AgentGym)	GPT-4-Turbo	AgentGym split	14.4

Sources: SwiftSage arXiv 2305.17390 (first 10 test variations/task = 270); ETO arXiv 2403.02502 / ACL 2024 (avg reward, seen/unseen); AgentGym arXiv 2406.04151. **Reading for us:** (1) a 7B open model with the right training (ETO 65–74, SFT 53–67) is a realistic target band for our Qwen3-4B rung-4 numbers — meaning 4B *can* get off the ground here, unlike the ASearcher worry. (2) SFT/BC is a *very* strong baseline on this env (BC-large 74.5, SFT-seen 67.4), so our SFT π_{base} floor rows matter enormously — an RL arm must clear a high bar, and AgentEvol *losing* to BC-large is a cautionary precedent. (3) The GiGPO/verl-agent and RAGEN lineages do **not** report ScienceWorld, so our rung-4 numbers are largely new territory for those exact methods — good for novelty, but no in-method reference point exists. **Caveat:** every follow-up uses a different eval subset (270 vs 194/241 vs AgentGym), so cross-row comparison is illustrative only; our SOTA table (§10 of the overview) will carry base-model + split + metric columns to avoid the mismatched-settings trap.

6 Challenges (why this rung matters for credit assignment)

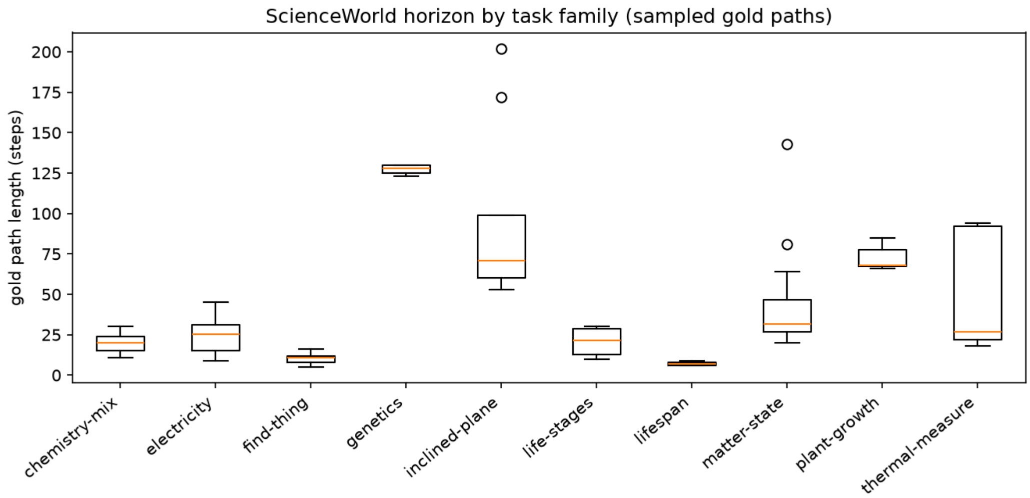
1. **Long, variable horizons (5–200+ steps).** Sampled gold-path lengths span ∼40× within one rung (see per-family table), giving RQ2 within-rung horizon strata for free — but also making this the hardest rung for flat trajectory-level credit, which is exactly the hypothesis under test.
2. **Combinatorial action space** (>100 valid combos/step) with soft failure on invalid actions — a free-text agent must *generate* legal actions, so a valid-action-rate metric is a leading diagnostic (as it was in the ALFWorld pilot).
3. **Partial observability + delayed dynamics.** Rewarding actions (focus, heat, wait) often pay off many steps later, and world state must be actively measured. This is the regime where a turn-level critic or anchor-state advantage should, in principle, beat token-broadcast GRPO.
4. **Catastrophic `focus` death (−100, instant end).** A per-turn hacking/instant-fail surface; needs a focus-death-rate counter and a decided terminal mapping.

5. **Heterogeneous dense reward (magnitudes 1–72, one subgoal often dominates).** Makes the naive "sum dense + terminal" channel spiky — the motivation for the redistribution-vs-separation side experiment.
6. **Context blow-up on long families.** Proposed 2×-gold turn caps reach 260–404 for genetics / matter-state / inclined-plane; full-history transcripts will not fit at 4B, so oldest-observation truncation is load-bearing for 4 of 10 families. Final caps await strong-model turns-to-success calibration.
7. **Task-sampling / variation imbalance (4 vs 692 variations).** Must be fixed a priori so "epoch" and the train/unseen generalization gap are well-defined.

Per-family horizons and proposed turn caps

family	sampled gold-len range	proposed cap (2×max)
lifespan	6–9	18
find-thing	5–16	32
chemistry-mix	11–30	60
life-stages	10–30	60
electricity	9–45	90
plant-growth	66–85	170
thermal-measure	18–94	188
genetics	123–130	260
matter-state	20–143	286
inclined-plane	53–202	404

Caps are provisional (2× the longest sampled gold path); they will be finalized from strong-model (gpt-oss-120b) rollout turns-to-success (95th percentile) rather than the gold heuristic alone, since gold paths contain filler an agent may skip or exceed. Annotated HTML trajectories in `assets/html/`.



Gold-path length (steps) by task family, 3 sampled train variations each — the ∼40× within-rung horizon spread that gives RQ2 its horizon strata.